

1. Administrative & Study Identification

Full Study Title: A Phase 1, Randomized, Double-Blind, Placebo-Controlled, Dose Escalation Trial to Evaluate the Safety, Tolerability, and Immunogenicity of V330 in Healthy Younger (18 to 49 Years Inclusive) and Healthy Older Participants (60 to 79 Years Inclusive) **Short Title/Acronym:** A Study of V330 in Healthy Younger and Healthy Older Participants (V330-001) **Protocol Number:** V330-001 **NCT Number:** NCT06630117 **Sponsor Name:** Merck Sharp & Dohme LLC **Investigational Product:** V330 Vaccine (Norovirus) **Phase of Development:** Phase 1 **Medical Monitor:** Merck Sharp & Dohme Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate whether a new vaccine, V330, is safe and well-tolerated in healthy younger and older adults. **Secondary Objectives:** To evaluate the immunogenicity of V330, specifically analyzing how well the body's immune system responds by producing antibodies after receiving varying dose levels of V330.

2.2 Background & Rationale To address the need for an effective prophylactic intervention against Norovirus infections. Previous studies on similar norovirus vaccines have demonstrated up to 61.8% effectiveness in preventing moderate to severe acute gastroenteritis caused by norovirus. This Phase 1 trial investigates the safety profile and the specific antibody responses generated by various dose levels of the multivalent V330 vaccine to determine optimal immunogenicity without compromising tolerability.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 20 to 100 volunteers **Number of Sites:** 4 locations in the United States (including centers in Florida and California) **Estimated Study Start:** 01-Mar-2024 **Estimated Primary Completion:** 23-Jul-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Is in good health prior to randomization. 2. Is an individual from 18 to 49 years of age inclusive, or from 60 to 79 years of age inclusive. 3. Ability to understand the purpose and risks of the study and provide signed and dated informed consent.

4.2 Exclusion Criteria 1. Has a history of myocarditis, pericarditis, and/or myopericarditis. 2. Has a history of cancer. 3. Current or previous therapies that may confound the assessment of vaccine safety or immune response.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Dose escalation of V330 administered at varying dose levels (e.g., DL 1 to DL 6) or placebo on Day 1, with specific cohorts receiving a second dose on Day 57. **Route of Administration:** Intramuscular (IM) injection. **Duration of Treatment:** Single or two-dose regimen up to Day 57, with follow-up monitoring extending until the end of the study.

5.2 Concomitant & Prohibited Therapy Permitted: General standard-of-care medications that do not interfere with immune response evaluation. **Prohibited:** Systemic immunosuppressants or therapies that could diminish or artificially alter vaccine immunogenicity.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Informed Consent, Medical History, Baseline Health Assessment
Baseline	Day 1	Randomization, First Dose (V330 or Placebo) Intramuscular Injection, Safety Monitoring
Interim	Day 57	Second Dose Intramuscular Injection (for specific cohorts), Safety Labs, Immunogenicity Assessment
End of Study	Follow-up	Final Vital Signs, Exit Interview, Final Immunogenicity / Antibody Titers, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Safety and tolerability of V330, defined by the frequency and severity of solicited injection site and systemic adverse events, as well as unsolicited adverse events post-vaccination. **Measurement Method:** Patient safety diaries and clinical/laboratory safety evaluations.

7.2 Secondary & Exploratory Endpoints 1. Immunogenicity: Change from baseline in norovirus-specific antibody titers (e.g.,

neutralizing antibodies) measured at designated post-vaccination time points.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Collection via participant diaries tracking injection site reactions (e.g., pain, swelling) and systemic symptoms (e.g., fever, fatigue). **Serious Adverse Events (SAEs):** Standard reporting timelines, typically within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Internal safety review committee overseeing dose escalation stages and overall trial safety.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all randomized participants who received at least one dose) and Per-Protocol / Immunogenicity Set (participants with evaluable antibody titer data). **Sample Size Justification:** Small sample size Phase 1 trial design (estimated $N = 20 - 100$) sufficient to evaluate basic safety and preliminary immunogenic signals rather than formal statistical power for efficacy. **Handling of Missing Data:** Missing serological data handled primarily through complete case analysis or Last Observation Carried Forward (LOCF) per standard early-phase vaccine protocols.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits to ensure proper vaccine storage, administration protocols, and safety diary compliance. **Audit Trail:** Robust electronic case report form (eCRF) locking and comprehensive data clarification processes.

11. Study Identifiers & Governance

Primary ID: V330-001 **Registry Number:** NCT06630117 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** A Study of V330 in Healthy Younger and Healthy Older Participants **Official Regulatory Title:** A Phase 1, Randomized, Double-Blind, Placebo-Controlled, Dose Escalation Trial to Evaluate the Safety, Tolerability, and Immunogenicity of V330 in Healthy Younger (18 to 49 Years Inclusive) and Healthy Older Participants (60 to 79 Years Inclusive)

12. Clinical Framework (Hyperkalemia Specific -> Adapted for V330 Vaccine)

Primary Condition: Healthy volunteers (Prophylaxis for Norovirus Infection) **Diagnosis Threshold:** N/A (Healthy profile baseline required) **Medical Methodology:** Prophylactic Vaccination **Optimization Target:** Achievement of protective norovirus-specific antibody responses

13. Study Procedures & Methodology

Management Pathway: Vaccine administration and staged safety observation/dose escalation **Education Protocol (HCP):** Site initiation and training on correct IM injection procedures and safety reporting **Education Protocol (Patient):** Onsite training on safety diary completion and side effect self-monitoring **Quality Audit Mechanism:** Clinical site monitoring and source data verification for safety and immunogenicity labs

14. Observation & Follow-Up Schedule

Total Duration: Follow-up tracking spans from Day 1 to estimated study completion in Jan-2027 **Onsite Visit Cadence:** Pre-defined safety checks post-Day 1 and Day 57 injections **Remote Monitoring Frequency:** Regular safety touchpoints as per Phase 1 protocol **Checklist Review Interval:** Review of patient-reported safety diaries continuously in the acute post-vaccination windows

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				